water. The formula describing this process, likewise relating to units of 1 mole, is:

$$H_2 + \tfrac{1}{2}O_2 \longrightarrow H_2O + 68{\cdot}4 \text{ kilocalories}$$

This means that the quantity of heat liberated is 68·4 kilocalories for every mole of water formed.

It may be said, quite generally, that these *heat emissions* of chemical processes are, throughout, of the order of magnitude of approximately 100 kilocalories per mole.

Now let us write the formula of a nuclear reaction, which happens to be one frequently applied in nuclear physics today, and which according to Döpel, can be released by ordinary canal rays having an energy corresponding to 5–10 kilovolts, viz.: the reciprocal reaction of two deuterons, which results in the formation of a hydrogen nucleus of mass number 1 and a hydrogen nucleus of mass number 3. The formula of the reaction, again relating to 1 mole, is:

$$_1D^2 + {}_1D^2 \rightarrow {}_1H^1 + {}_1H^3 + 100{,}000{,}000 \text{ kilocalories}$$

In other words, the two deuterons do combine at first, but the compound nucleus thus formed splits up immediately into the two nuclei indicated above, since this is more favourable from the point of view of energetics, as is demonstrated by the tremendous heat emission of 100,000,000 kilocalories per mole. The heat emissions in the majority of the other nuclear reactions are of a similar order of magnitude. We see, therefore, how important nuclear reactions become as soon as they can be produced on a really large scale. The energy yield of nuclear reactions is about one million times greater than that obtainable by chemical processes. In other words, the nuclear ‘combustion’ of matter yields a millionfold of the energy released by a chemical combustion of the same quantity of the substance.

In nature, such nuclear reactions occur, as a rule, on a small scale only—due to the effect of cosmic radiation and of the radiation of radioactive substances. The fact that the energy released by these natural phenomena is not observed, is due to the fact that the quantities of energy produced are far too small, and also far too scattered.

Nevertheless, it is not quite right to claim that the energy liberated by nuclear reactions has never before played any part in nature. On the contrary,